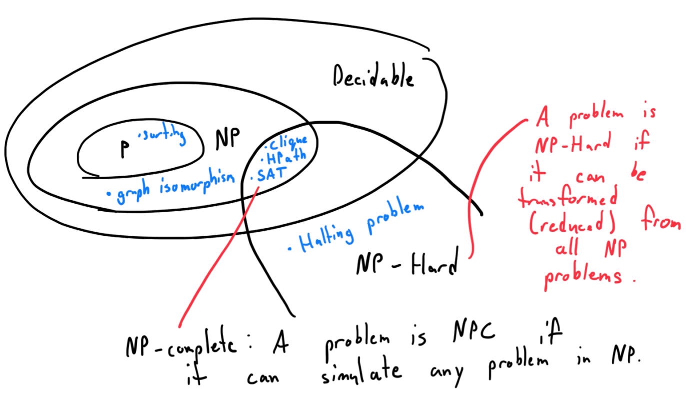
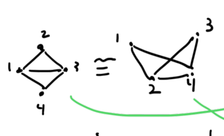
Reductions

$P = NP$ or $P \neq NP$?



NP-complete: A problem is NPC if it can simulate any problem in NP.

Graph isomorphism:



$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 0 & 1 & 0 \end{array} \qquad \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 1 & 1 \\ 3 & 0 & 1 & 0 & 1 \\ 4 & 1 & 1 & 1 & 0 \end{array}$$

adjacency matrices

Are the two graphs actually the same with different vertex labels?
(i.e. do their adjacency matrices give the same graph drawing?)

Note: If problem can be solved in polynomial-time on a nondeterministic TM (i.e. it's in NP) then immediately any solution can be verified quickly on a deterministic TM. This is obvious. The computation tree of the NTM has polynomial-depth and a standard computer can easily check if a solution is valid since all the correct branches are already given.

Reductions

Satifiability Problem:
SAT $= \{\langle\psi\rangle \mid \psi$ is a satisfiable boolean expression$\}$.

$$\psi = (\underbrace{\overset{T}{\cancel{\bar{x}}} \vee \overset{T}{\cancel{y}}}_{\text{clause}}) \wedge (\overset{F}{\cancel{x}} \vee \overset{T}{\cancel{\bar{z}}})$$

literals (true or false)

$x = F$
$y = T$
$z = F$

$$(T \vee T) \wedge (F \vee T)$$
$$(T) \wedge (T)$$
$$\psi = T$$

SAT was the first NPC problem.
3SAT: Each clause has 3 literals.

Recall CLIQUE: Does there exist a clique of size k in a given graph.

$\boxed{\text{Theorem 7.32}}$ 3SAT is polynomial-time reducible to CLIQUE.

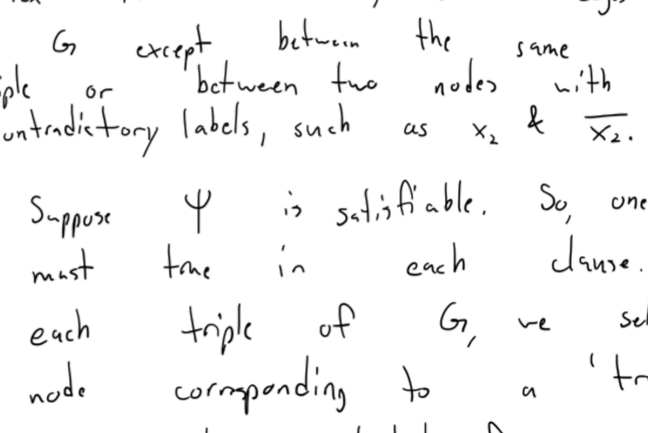
Proof: Let ψ be a SAT expression with k clauses. Organize k triples, each corresponding to a clause in ψ, to form graph G.

$$\psi = (a_1 \vee b_1 \vee c_1) \wedge (a_2 \vee b_2 \vee c_2) \wedge \dots \wedge (a_k \vee b_k \vee c_k).$$

Ex: if $\psi = (x_1 \vee x_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee \bar{x}_2) \wedge (\bar{x}_1 \vee x_2 \vee x_2)$

We will transform every instance of 3SAT to some instance of CLIQUE.

Here is an example reduction:

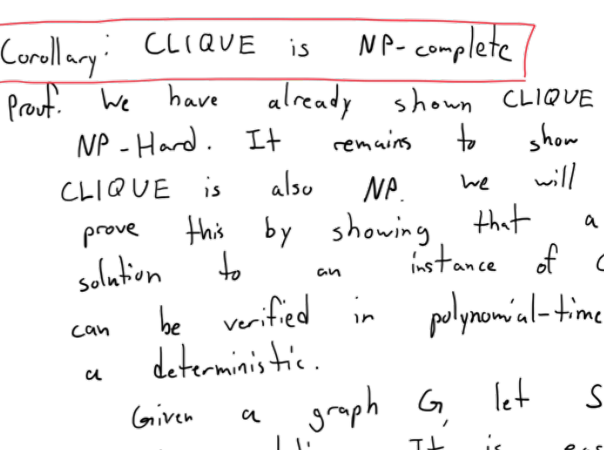


Each literal in ψ has a vertex in G. Now, add all edges to G except between the same triple or between two nodes with contradictory labels, such as x_2 & $\bar{x}_2$.

$\Rightarrow$ Suppose ψ is satisfiable. So, one literal must true in each clause. In each triple of G, we select one node corresponding to a 'true' literal. The nodes selected form a k-clique.

$\Leftarrow$ Suppose G has a k-clique. No two vertices of the clique appear in the same triple. We make each literal corresponding to the vertices true. Thus, ψ is satisfiable. □

We have shown CLIQUE is NP-Hard.



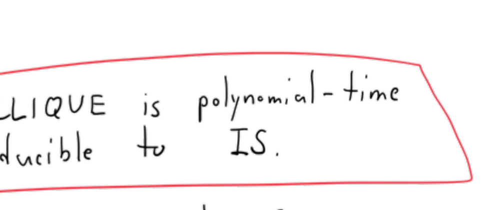
- A polynomial-time algorithm to CLIQUE would immediately imply a poly-time solution to 3SAT.

$\boxed{\text{Corollary: CLIQUE is NP-complete}}$

Proof: We have already shown CLIQUE is NP-Hard. It remains to show CLIQUE is also NP. We will prove this by showing that a solution to an instance of CLIQUE can be verified in polynomial-time via a deterministic.

Given a graph G, let $S \subseteq G$ be a given solution. It is easy to iterate through the vertices of S and check that all pairs are adjacent and $|S| = k$. □

Independent Set: A subgraph of a graph G s.t. each pair of vertices is not adjacent.



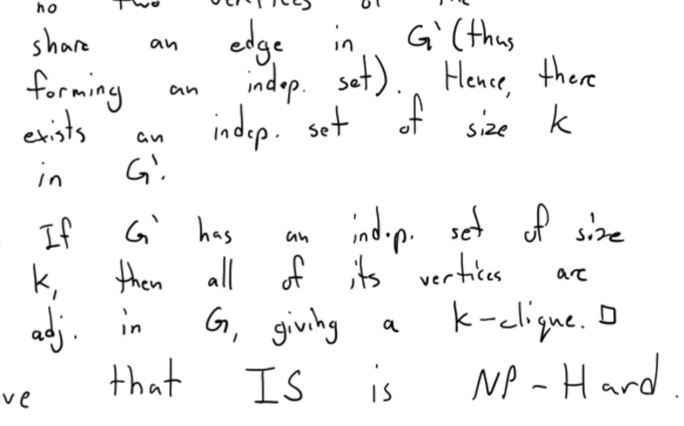
IS (problem): Does there exist an indep. set of size k in a graph G?

$k = 4$?

$\boxed{\text{Theorem: CLIQUE is polynomial-time reducible to IS.}}$

Proof: Given a graph G, we construct $G' = (V', E')$ s.t. $V' = V$ and $E' = \bar{E}$.

Ex:



$\Rightarrow$ If G has a k-clique, then no two vertices of the set share an edge in G' (thus forming an indep. set). Hence, there exists an indep. set of size k in G'.

$\Leftarrow$ If G' has an indep. set of size k, then all of its vertices are adj. in G, giving a k-clique. □

We have that IS is NP-Hard.

$\boxed{\text{Corollary: IS is NPC.}}$

Proof: Given a graph $G = (V, E)$, a solution to IS $H \subseteq G$, it can be verified that H is an indep. set in $O((|V| + |E|)^2)$ by iterating through the pairs of vertices in H and checking that they are all not adjacent. □
Since IS is in NP and NP-Hard, it is NP-complete. □
